

# How Constant Is Your Beat?

Computer-Assisted Analysis of Beat and Tempo Fluctuations from Acousmatic Music to Minimal Techno with “beat\_it” Toolbox.

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## **Rhythm under the Microscope**

An Interdisciplinary Conference on  
Microrhythm and Groove in Popular Music

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# What is „beat\_it“?

- A toolbox primarily designed for various types of pitch and onset detection, as well as rhythmic grid analysis.
- It utilizes a range of statistical and machine learning algorithms, combined with powerful visualization and analysis libraries such as librosa and bokeh.
- Tailored for qualitative analysis from artistic and music-theoretical perspectives.



<= Colab

GitHub =>



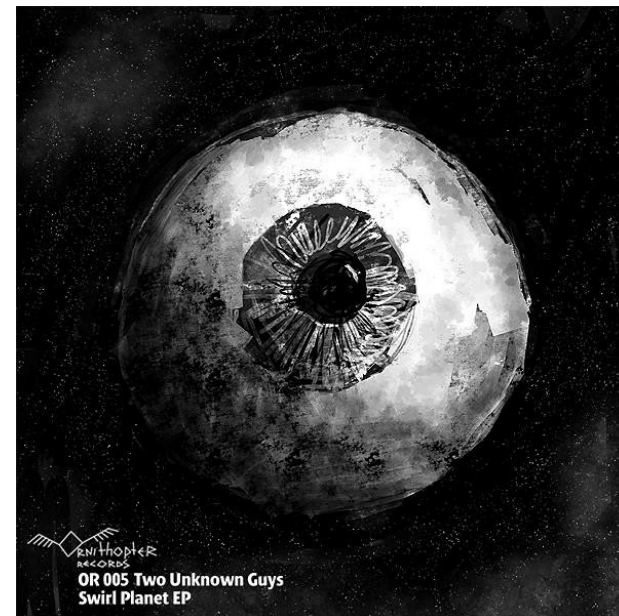
# Workshop Overview

- Personal Background
- Analyzing electroacoustic music
  - Denis Dufour - *Bocalises - Prélude* (1977-1978)
  - Bernard Parmegiani - *Dedans-Dehors - jeux* (1977)
- Analyzing minimal techno
  - Tekonivel (Miika Vainio) – *Sirkus – Voimamies* (1996)
  - The Vision (Robert Hood) – *Spectral Nomad* (1996)
- Q&A

# Personal Background

- Began studying piano and composition in 2004 at HMT Leipzig.
- Gradually became more involved in the local electronic music scene—DJing, making friends on early 2000s music forums (like [techhouse.ee](http://techhouse.ee)).
- Slowly transitioned to producing minimal and techdub, while still composing "traditional" instrumental music at HMT.
- **Faced resistance:** most of my professors had little to no understanding or appreciation for techno music.
- **Led to an inner conflict:** How can I justify my love for techno music? Through analysis? How can I express the elements that fascinate me?

- **2007-2010:** Peak involvement in the techdub/minimal scene
  - Co-founded *Ornithopter Records* with Christoph Schindling and Ilya Zonov.
  - Launched techdub/minimal project *Two Unknown Guys* with Ilya Zonov.
  - Played DJ/live sets at raves and underground clubs in and around Leipzig.



- **2010: Setback**
  - Just as the label started making money, Christoph Schindling squandered it on drugs and disappeared for months. Thankfully, he later took responsibility and tried to make things right with the artists.
- **2010-2011:** Attended composition master's program at HMDK Stuttgart under Marco Stroppa as well as my Meisterklasse in Electroacoustic Music under Ipke Starke at HMT Leipzig
- **2012:** Doctoral studies (electroacoustic music/musicology/music theory) at HMT Leipzig → academic career prevailed
- **2018:** Earned doctorate, got postdoc research position, got family and kids.
  - No longer actively participating in the techno music scene, but still collecting a *lot* of vinyl.

- Since studying with Marco Stroppa, I developed a strong interest in algorithmic composition, mastering Max, Audiosculpt, and OpenMusic.
- Since the late **2010s**, I gradually transitioned into Python programming (change of focus from creating to analyzing).
- From **2021-22**, I worked as a researcher on the CAMAT project at HfM Weimar, led by Martin Pfeleiderer, where we developed a Python-based toolbox for symbolic music analysis.
- During this time, I got familiar through my colleague Christon-Ragavan Nadar with FMP notebooks and various machine learning topics.
- Gradually, I created several Python tools for analyzing rhythmic and spectral features, leading to the development of the **beat\_it** toolbox.

# Spectromorphology

- Introduced in 1997 by Dennis Smalley as an analytical theoretical framework for electroacoustic music
- Used to describe and interpret the ways in which sounds are shaped and interact within a musical context
- Particularly useful for analyzing complex, non-traditional musical forms, such as acousmatic and electroacoustic compositions, where traditional music theory may not fully apply
- Relies on a specialized **vocabulary to describe the characteristics and behaviors of sound**
- Focuses on the perceptual aspects of how sound evolves and interacts within a piece of music
- Key categories of descriptive terms in spectromorphology
  - Spectral Terms
  - Morphological Terms
  - Behavioral Terms
  - Source-Cause Terms

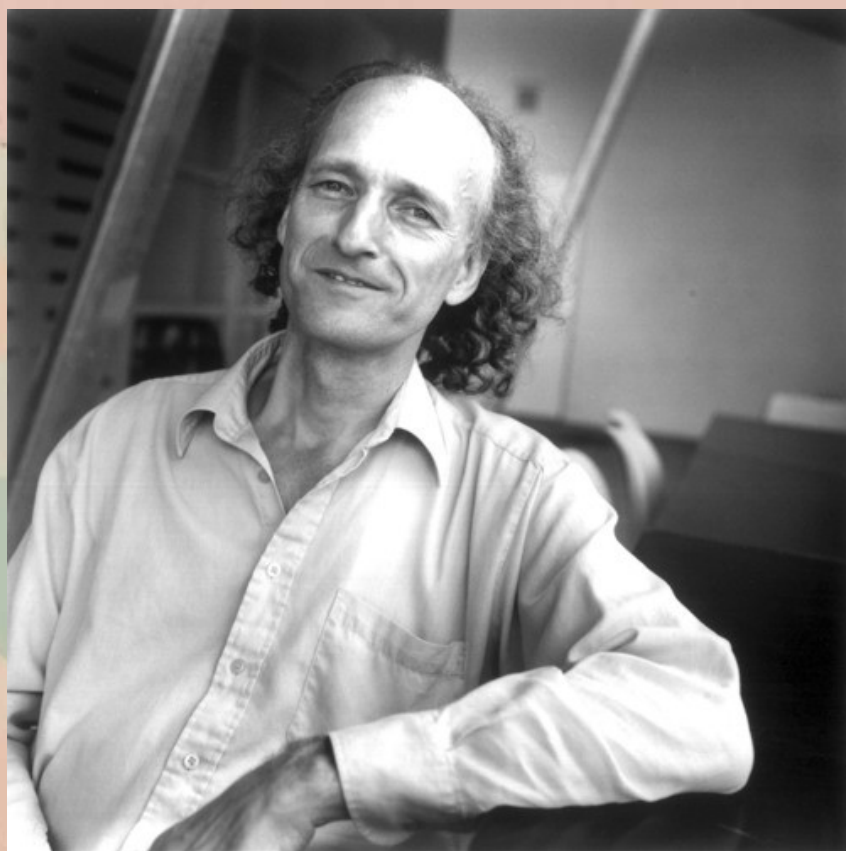
- **IGNORING TECHNOLOGY** (Smalley 1997)

*“The sonogram, a graphic spectral analysis by computer, has been regarded as a solution to the visual representation of electroacoustic music. I regard it as a very useful aid rather than a solution. A sonogram is a type of literal spectral analysis at a chosen visual resolution: at too high a resolution, detail becomes lost in a blur; at too low a resolution, there is insufficient detail.”*

*“In spectromorphological thinking, we must try to ignore the electroacoustic and computer technology used in the music's making. Surrendering the natural desire to uncover the mysteries of electroacoustic sound-making is a difficult but necessary and logical sacrifice.”*

# Denis Dufour - Bocalises – Prélude (1977-78)

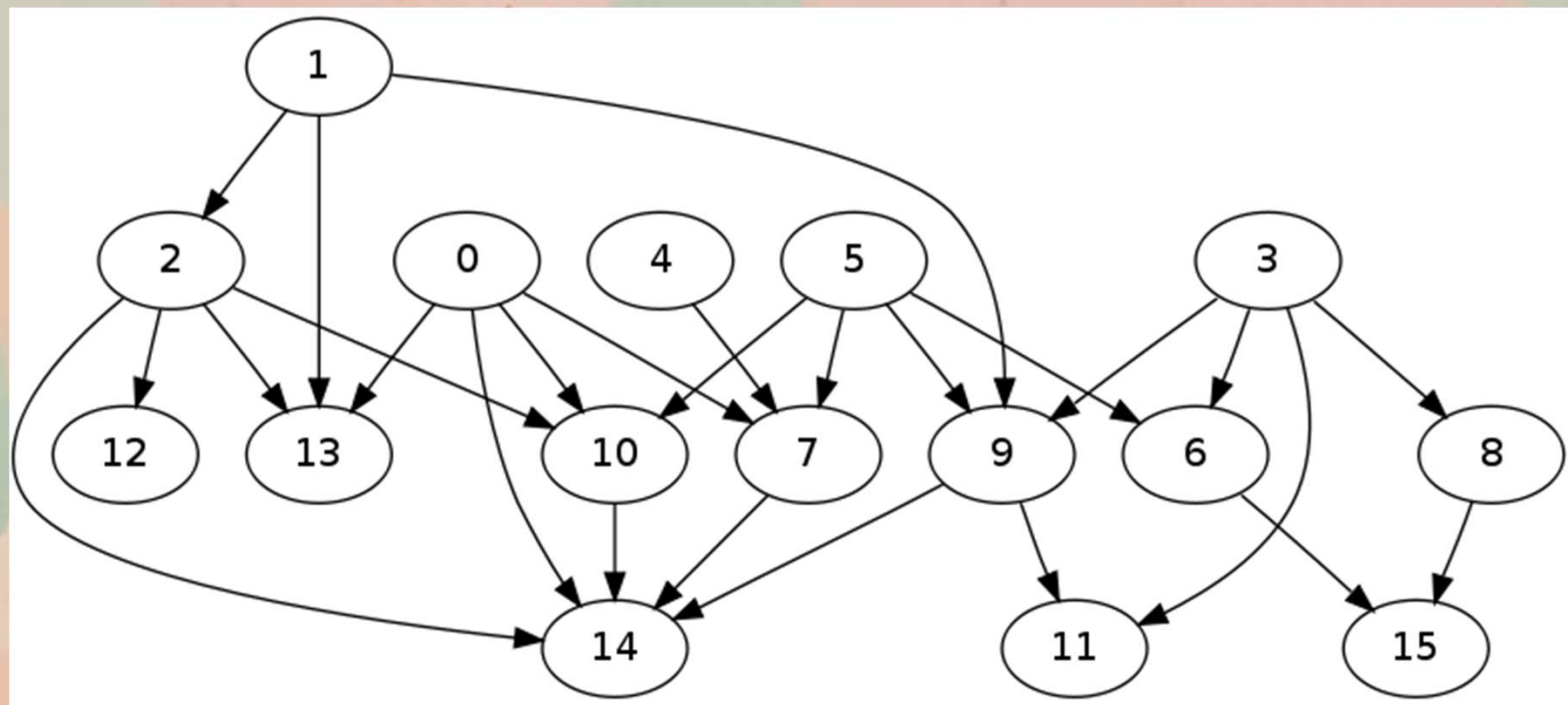
- **Denis Dufour** (born October 9, 1953, in Lyon) is a renowned French composer known for his work in instrumental, electroacoustic, and acousmatic music.
- Studied at the Lyon Conservatory and the Paris Conservatory, where he worked with Pierre Schaeffer and Guy Reibel.
- A prominent figure in the Groupe de Recherches Musicales (GRM).
- With over 180 works, Dufour has contributed extensively to contemporary music by blending instrumental, electroacoustic, and live electronic techniques.
- Bocalises – Limited information is available about the compositional process, but the sound material was obtained through various manipulations of glass jars.



- **dufour\_peaks\_manual.ipynb**: Basic cue points and IOI (Inter-Onset Interval) analysis, with statistical evaluation.
- **dufour\_onsets\_librosa.ipynb**: Automated onset extraction using librosa and onset filtering through Directed Acyclic Graph (DAG) methods.
- **dufour\_csv\_onset\_filter.ipynb**: General onset filtering/extraction using DAG, with accuracy evaluation metrics.
- **dufour\_pulsation.ipynb**: Rhythmic grid calculations, modeling, and optimization techniques.
- **dufour\_features\_bokeh.ipynb**: Audio feature extraction and visualization using librosa and bokeh.

# Directed Acyclic Graph (DAG)

- A **Directed Acyclic Graph (DAG)** is a type of structure used in computer science and statistics that is quite useful for representing systems where certain events or actions lead to others in a one-way, non-repeating fashion.
- **Directed** means that the connections between the events (or "nodes") have a direction. For example, if we think about tasks in a workflow, one task might need to happen before the next one. The arrow between two tasks (nodes) shows the direction in which the events are related.
- **Acyclic** means that once you move in one direction, you cannot go back. In other words, you can't create loops or cycles. For instance, if you start from Task A, move to Task B, and then to Task C, you cannot circle back to Task A. Everything progresses forward without repeating.



# Bernard Parmegiani - Dedans-Dehors - jeux (1977)

- **Bernard Parmegiani** (1927 – 2013) is a pioneer and leading light of electroacoustic music. He leaves behind him a vast and fascinating œuvre, a priceless heritage for today's generation.
- His life was structured around a few mainstays: working in a studio filled with machines, lighting a pipe filled with Amsterdamer, in the company of his wife (his “first ear”), and not least, a few affectionate cats.

(<https://maison-ona.com>)

- ‘Dedans-Dehors’ is a direct sequel to ‘De Natura Sonorum’. After having used different compositional writing styles (letting the notion of sounds’ “nature” appear through sounds originating from all different types of things, especially in ‘De Natura Sonorum’), there still remained the ever-abundant world of the noises that result from human activity, which I’ll call “artificial” so as not to confuse with “natural”. These noises are the clue to life which is, whether close or far, physical or temporal, in the present. Natural sounds possess a power that speak to us at different levels. They reveal a force, an energetic dynamism, which escapes us and yet which we attempt to grab a hold of through the bias of the internalized act of listening. Being able to record them facilitates a progressive logic to their existence. Through our interpretations and our expressions when listening to sonic material, we initiate a metamorphosis from the interior to the exterior. The notion of metamorphosis, or transformation, is one of the principles that directs the shape of this set of movements.

(Bernard Parmegiani)



- **parm\_librosa\_onsets.ipynb**: Reliable detection of water drop sequences using DAG onset filtering.
- **parm\_pulsation.ipynb**: Rhythmic grid separation and alignment techniques.
- **parm\_pitch.ipynb**: Methods for automatic pitch detection.
- **parm\_pitch\_eval.ipynb**: Evaluation of pitch detection accuracy.

# Tekonivel – Sirkus – Voimamies (1996)

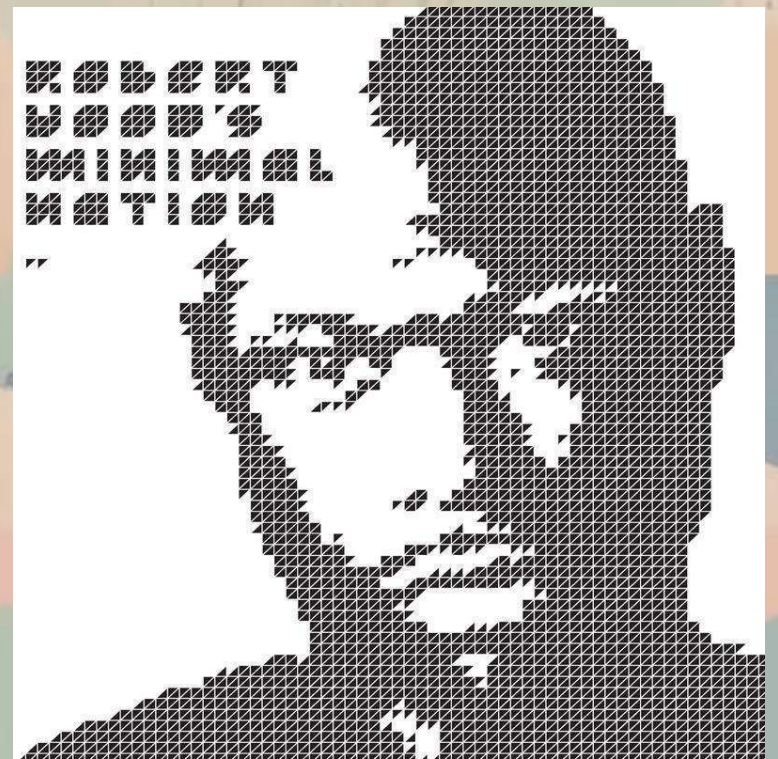
- **Miika Vainio** (1963 -2017) was a visionary Finnish electronic musician and composer, celebrated for his groundbreaking work in the IDM and experimental electronic genres.
- As a co-founder of Sahkö Records, Vainio was instrumental in establishing the label as a prominent platform for avant-garde and innovative electronic music.
- Famous for his Ø and Pan Sonic (together with Ilpo Väisänen and Sami Salo) projects



- **tekonivel\_onsets.ipynb**: Challenges in onset detection due to overlapping synthesizer and bass drum sounds.
- **tekonivel\_features\_bokeh.ipynb**: Generation of onsets based on rhythmic grid projections.
- **tekonivel\_features\_bokeh2.ipynb**: Rhythmic grid projections as a general analysis method.

# The Vision – Spectral Nomad (1996)

- **Robert Hood** (born 1965) is an American electronic music producer and DJ. He is a founding member of the group Underground Resistance as a 'Minister of Information' with Mad Mike Banks and Jeff Mills. He is often considered to be one of the founders of minimal techno.
- The Vision is an early alias of Robert Hood.
- He released his first record, "Gyroscopic EP", on the Underground Resistance label in 1991 and also released as The Vision on Tresor, Metroplex and his own Hardwax label.



- **vision\_librosa\_onsets.ipynb**: Assessing the accuracy of onset detection.
- **vision\_librosa\_onsets2.ipynb**: Maximizing onset detection sensitivity.
- **vision\_pulsation.ipynb**: Saddest base value minima ever. ☹️
- **vision\_features\_bokeh.ipynb**: Investigating looped structures.
- **vision\_features\_bokeh2.ipynb**: Achieving beat-precise segmentation and similarity detection.

## Some closing thoughts?

- Through an analytical approach to minimal techno, I discovered complexities that I had not previously encountered in traditional electroacoustic music.
- The intricate details within minimal techno can be overwhelming and challenging to classify or dissect. However, it becomes highly inspiring when you can analytically outline how certain looped structures function especially if it's a very simple algorithm/model behind.
- During my analysis and tool refinement, I found it particularly insightful that using similar analytic tools allows you to view both electroacoustic music and minimal techno not as entirely distinct genres, but as similar analytic domains with a surprising amount of commonality.

# Thank you!

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<https://egorpol.github.io/>

[https://github.com/egorpol/beat\\_it](https://github.com/egorpol/beat_it)

